



Compiler Construction

Lab 1

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Task 1:

Code:

```
//task1.We love compiler
#include <iostream>
using namespace std;
int main()
{
    string input;
    cout<<"Enter sentence:";
    getline(cin,input);
    string token = "";
    for(int i =0;i<input.length();i++)
    {
        if(input[i]!=' ')
        {
            token +=input[i];
        }
        else
        {
            cout<<"Token: "<<token <<endl;
            token = "";
        }
    }
    cout<<"Token: "<<token <<endl;
    return 0;}
```

Output:

```
C:\Users\Riphah\Desktop\T1.€ X + v
Enter sentence:We Love Compiler
Token: We
Token: Love
Token: Compiler

-----
Process exited after 7.072 seconds with return value 0
Press any key to continue . . . |
```

Task 2:

Code:

```
//task2. int total = sum + 10;
#include <iostream>
using namespace std;
int main()
{
    string input;
    cout<<"Enter statement: \n";
    getline(cin, input);
    string token = "";
    for(int i=0;i<input.length();i++)
        {if(input[i]!=' ')
            {
                token +=input[i];
            }
            else
            {
                if(token=="int"||token=="float")
                {
                    cout<<token<<": KEYWORD\n";
                }
            }
        }
```

```

else if (token=="+"||token=="=")
    {
        cout<<token<<": OPERATOR\n";
    }

    else if(token[0]>='0'&&token[0]<='9')
    {
        cout<<token<<": NUMBER\n";
    }
else if(token==";")
{
    cout<<token<<": TERMINATOR\n";
}

else
{
    cout<<token<<": IDENTIFIER\n";
}
token="";
}
}

if(token=="int"||token=="float")
{
    cout<<token<<": KEYWORD\n";
}
}

else if (token=="+"||token=="=")
    {
        cout<<token<<": OPERATOR\n";
    }

    else if(token[0]>='0'&&token[0]<='9')

```

```

        {
            cout<<token<<": NUMBER\n";
        }
    else if(token=="")
    {
        cout<<token<<": TERMINATOR\n";
    }

    else
    {
        cout<<token<<": IDENTIFIER\n";
    }

    return 0;
}

```

Output:

```

C:\Users\Riphah\Desktop\t2.ε
Enter statement:
int total sum = 10 ;
int: KEYWORD
total: IDENTIFIER
sum: IDENTIFIER
=: OPERATOR
10: NUMBER
;: TERMINATOR

-----
Process exited after 7.624 seconds with return value 0
Press any key to continue . . .

```

Task 3:

Code:

```
//task3. recognize if, while, for
```

```
#include <iostream>
```

```

using namespace std;

int main()
{
    string input;
    cout<<"Enter statement: \n";
    getline(cin, input);
    string token = "";
    for(int i=0;i<input.length();i++)
    {
        if(input[i]!=' ')
        {
            token +=input[i];
        }
        else
        {
            if(token=="int"||token=="float")
            {
                cout<<token<<": KEYWORD\n";
            }
            else if (token=="+"||token=="="||token=="<"||token==">")
            {
                cout<<token<<": OPERATOR\n";
            }
            else if(token[0]>='0'&&token[0]<='9')
            {
                cout<<token<<": NUMBER\n";
            }
            else if(token=="")
            {

```

```

        cout<<token<<": TERMINATOR\n";
    }
    else if(token=="while"||token=="for")
    {
        cout<<token<<": LOOP STATEMENT\n";
    }
    else if(token=="if")
    {
        cout<<token <<": CONDITIONAL STATEMENT\n";
    }
    else if(token=="("||token=="")||token=="["||token==""]||token=="{"||token=="}")
    {
        cout<<token <<": PARENTHESES\n";
    }

    else
    {
        cout<<token<<": IDENTIFIER\n";

    }
    token="";
}

    if(token=="int"||token=="float")
    {
        cout<<token<<": KEYWORD\n";
    }
    else if (token=="+"||token=="="||token=="<"||token==">")

```

```

    {
        cout<<token<<": OPERATOR\n";
    }
    else if(token[0]>='0'&&token[0]<='9')
    {
        cout<<token<<": NUMBER\n";
    }
    else if(token=="")
    {
        cout<<token<<": TERMINATOR\n";
    }
    else if(token=="if"||token=="while"||token=="for")
    {
        cout<<token<<": LOOP OR IF STATEMENT\n";
    }
    else if(token=="if")
    {
        cout<<token <<": CONDITIONAL STATEMENT\n";
    }
    else if(token=="("||token==")"||token=="["||token=="]"||token=="{"||token=="}")
    {
        cout<<token <<": PARENTHESSES\n";
    }
    else
    {
        cout<<token<<": IDENTIFIER\n";
    }

    return 0;

```

}

Output:

```
C:\Users\Riphah\Desktop\t3.ε X + v
Enter statement:
for ( int i = 0 ; i < 7 ; i + + )
for: LOOP STATEMENT
(: PARENTHESES
int: KEYWORD
i: IDENTIFIER
=: OPERATOR
0: NUMBER
;: TERMINATOR
i: IDENTIFIER
<: OPERATOR
7: NUMBER
;: TERMINATOR
i: IDENTIFIER
+: OPERATOR
+: OPERATOR
): PARENTHESES

-----
Process exited after 32.59 seconds with return value 0
Press any key to continue . . .
```